

ARK-455b Results

ProofRecord Verification After Tampering — Corrected Retest with Validity-Window Semantics

Verdict:  PASS

Execution date: 2026-07-17

Substrate: Local classical software (no IBM QPU)

Lock commit: 6776616 (prereg + code + MANIFEST committed before execution)

Kill-gate: PASS (100.0% V1-V2 concordance; both verifiers accepted all 50 valid records; 50/50 tampered carried effective mutations)

1. Summary

ARK-455b is the corrected retest of ARK-455, whose v1.0 FAIL was traced to a test-harness no-op in the Arm-3 timestamp tamper (see [../ark-455/CORRECTION.md](#) (`../ark-455/CORRECTION.md`)). With the tamper corrected and a validity-window gate added, **all three preregistered criteria pass with perfect margins**, and the two new-in-455b conditions behave exactly as the doctrine predicts:

- **Arm 3 (timestamp, real post-signing +1s):** now detected at **100%** by both verifiers — directly confirming that the ARK-455 0% “detection failure” was an artifact of a no-op mutation, not a verifier or specification defect.
- **Arm 9 (timestamp, pre-signing expired, valid signature):** rejected at **100%** by the validity-window gate — a record that a bare signature check (ARK-455) would have accepted.

Metric	Value	Threshold	Result
C1 — V_accept_original (Arm 1)	1.0000	≥ 0.95	✓
C2 — V_reject_min (Arms 2-9)	1.0000	≥ 0.95	✓
C3 — V_margin	1.0000	≥ 0.85	✓
V1-V2 concordance (all arms)	100.00% (900/900)	$\geq 99\%$	✓
Mutation-effectiveness (Arms 2-9)	100% (800/800)	100%	✓

2. Design as Executed

- **9 arms × 100 records = 900 records** (the concrete design registered in PREREGISTRATION.md; no other variant was run).
- **Dual independent verifiers:** V1 (JavaScript, tweetnacl), V2 (Python, PyNaCl).
- **Two-gate verification:** Gate A = Ed25519 signature over RFC 8785 (JCS) canonical form of the 7 signed fields; Gate B = validity window (reject future-dated `age < 0` or expired `age > TTL`). **TTL = 300 s.**
- **verification_time** captured after record generation (auditor verifies an issued record), modelling realistic issuance→verification ordering.
- **Mutation-effectiveness gate:** every REJECT-arm record was confirmed to carry an effective mutation before scoring; the run would have aborted otherwise.

3. Per-Arm Results

Arm	Label	Mode	V1 acc / rej	V2 acc / rej	Mut-eff	Concordance
1	ACCEPT-original	control	100 / 0	100 / 0	n/a	100/100
2	REJECT-decision	post-sign	0 / 100	0 / 100	100/100	100/100
3	REJECT-timestamp-postsign	post-sign (real +1s)	0 / 100	0 / 100	100/100	100/100
4	REJECT-payload_hash	post-sign	0 / 100	0 / 100	100/100	100/100
5	REJECT-evidence_refs	post-sign	0 / 100	0 / 100	100/100	100/100
6	REJECT-actor	post-sign	0 / 100	0 / 100	100/100	100/100
7	REJECT-outcome	post-sign	0 / 100	0 / 100	100/100	100/100
8	REJECT-review_path	post-sign	0 / 100	0 / 100	100/100	100/100
9	REJECT-timestamp-presign-expired	pre-sign (valid sig, expired)	0 / 100	0 / 100	100/100	100/100

Gate attribution: Arms 2-8 rejected via Gate A (broken signature). Arm 9 rejected via Gate B (valid signature, out-of-window timestamp) — the condition Gate A alone cannot catch.

4. Recorded Seeds (256-bit)

Kill-gate calibration seed:

74286458217247489335822290487316412859199419933419266139920589335769580693475

Kill-gate verification_time: 2026-07-17T18:04:42.222404Z

Arm run window: start 2026-07-17T18:04:50.268672Z → end 2026-07-17T18:05:39.138979Z

Arm	Seed
1	303387027884101712794861083672595012 844429997070352354678144450627618443 86588
2	114658493849469925467984076741683536 542222061775061141230437307184350924 665141
3	105395414605302171008035296597720735 394719011550540308140847548311148061 74598
4	796131727100843600171503269569938280 221261912437968390231548297592394041 00405
5	316574116376001928493724842301159488 892650322136297033823753136550832378 93451
6	482984885354321554297884609048746411 176463756069933548849507201340182536 90161
7	127858980930609150039406398180536308 323976094252538206019363125648885815 85415
8	479400870783548319430392003808956841 120778479026441156901817248523106403 18887
9	134125076914969490812602636303186943 316123947973121343735580274355052971 24233

Full per-arm data in `results/arm_1_results.json` ... `results/arm_9_results.json` ;
kill-gate detail in `results/killgate_calibration.json` ; overall metrics in
`results/overall_results.json` .

5. Interpretation

H1 (primary) — CONFIRMED. Independent verifiers accept genuine in-window originals and reject every tampered and every expired record. All of C1-C3 pass at 1.0000.

H2a (tampering universality) — CONFIRMED. All seven post-signing tampering targets, including the corrected timestamp arm, are detected at 100%. No target is exempt.

H2b (dual verifier agreement) — CONFIRMED. 100.00% concordance across all 900 records (independent JS and Python implementations).

H2c (signature binding) — CONFIRMED. No post-signing tampered record (Arms 2–8) produced a valid signature.

H2d (validity-window binding) — CONFIRMED. Arm 9 records, whose signatures are cryptographically valid, are rejected at 100% because their timestamps are out of the 300 s window. Temporal validity is enforced independently of signature validity (expiry doctrine of ARK-442, now at the verification layer).

H2e (mutation effectiveness) — CONFIRMED. 800/800 REJECT-arm records carried an effective mutation. The class of defect that produced the ARK-455 anomaly cannot pass this harness silently.

Relationship to ARK-455

ARK-455b does **not** retract ARK-455’s recorded FAIL. It (1) confirms the erratum’s root-cause finding by showing that a real timestamp tamper is detected at 100%, and (2) extends the specification with validity-window semantics. The ARK-455 FAIL stands as an honest record of a harness defect; the capability it appeared to question is demonstrated intact here.

6. Honest Boundary Statements

- Classical software test; no quantum claim.
- Single-field tampering only (one mutation per record); multi-field tampering, replay across keys, and adversarial signature forgery remain deferred.
- A PASS here validates the verification model and expiry semantics under the registered arms; it does not claim coverage beyond them.
- Results are reported exactly as produced by the locked code and recorded seeds.

Remnant Fieldworks Inc. — ExecutionProof ARK Series

Repository: <https://github.com/derekhone/executionproof-testbeds>